

SENTINEL

Statewide CCTV Integration Platform & ANPR Surveillance Engine

- **Event:** Gujarat CCTV Integration Hackathon 2026
- **Category:** Category 1 (Academic / Research / Startup)
- **Team Name:** KodeMatrix
- **Track Scope:** Model 1 (Central CCTV Registry & GIS Mapping) + Model 2 (Unified Viewing Platform & ANPR Alerting)

A production-grade platform running against the live Sentinel sandbox grid: 30 cameras onboarded and mapped, a continuous ANPR pipeline validated at 100% on legible footage, real-time watchlist alerting, and timestamped route reconstruction across cameras.

Slide 2: The Problem and What Wins It

The Operational Gap in Statewide Surveillance

- Cameras are owned by multiple departments in diverse formats with no central registry.
- Investigators asking "where has this vehicle been?" have no unified system to query across jurisdictions.
- Watchlist matching, where present, relies on manual human monitoring across isolated displays.

What Sentinel Delivers

Core Capability	Operational Functionality
Central Onboarding	Ingests any camera published in the catalogue with location provenance recorded
Unified Viewing	Live multi-camera viewing wall without per-department client software
Continuous ANPR	Automatic plate indexing: every plate, every camera, every timestamp
Watchlist Matching	Instant exact-then-fuzzy trigram matching on every detection
Real-Time Alerts	Low-latency WebSocket alert dispatch with watchlist severity levels
Route Reconstruction	Interactive map timeline of vehicle sightings with speed estimation
Complete Auditability	Purpose-bound audit logging for every citizen search and evidence export

Slide 3: System Architecture



Technology Stack: Python 3.12+ / FastAPI | PostgreSQL 17 + PostGIS + pg_trgm | React + Vite + TypeScript | Leaflet | hls.js | ONNX Runtime (CPU)

Architectural Discipline: Kafka, Elasticsearch, and Kubernetes were deliberately omitted at this scale because PostgreSQL handles all indexing cleanly. The scale-out path for 80,000 cameras is fully designed and documented in Section 9 of the HLD.

Slide 4: Model 1 - Registry & GIS Mapping

Handling Real-World Data Gaps with Integrity

The sandbox catalogue publishes **only** `id` and `name` : no coordinates, no department, and no codec hints.

Our Solution

- Resolved all 30 camera locations from real site names published in the catalogue.
- Implemented strict location provenance precedence: hand-verified → Nominatim geocoding (rate-limited, cached, bounded to Gujarat) → district centroid → **null**.
- Stored `geo_source` and `geo_confidence` on every camera record so provenance travels into reports and UI displays.

What We Refused To Do

We did not invent arbitrary coordinates. An initial arithmetic grid approach placed Junagadh cameras in Surat; that was removed. Unlocatable cameras are reported transparently as unlocated.

GIS Capabilities: PostGIS geography points, GeoJSON export, department / status / confidence map layers, clustered markers, and click-through to live video tiles.

Slide 5: Model 2 - ANPR & Accuracy Engine

Local CPU Inference | Zero Third-Party API Dependence

Pretrained open-source models (YOLOv9-t detector, cct-s-v2 OCR) executing on **local CPU**. No video frame ever leaves the deployment; no external service sits in the alerting path.

Accuracy Achieved Through Domain-Specific Engineering

1. Track-Level Voting: Vehicles remain visible across 20-60 frames. Every read votes per character weighted by OCR confidence, right-aligned on the 4-digit serial. Tracks reach 20-35 reads and accumulate 1.000 confidence.

2. Indian Plate Grammar: [2 letters][1-2 digits][1-3 letters][4 digits] . Position rules make O↔0, I↔1, S↔5, B↔8, Z↔2, G↔6 deterministically correctable, with state codes validated against RTO lists.

Real Example: OCR returned GJ01AB1234 . Position 2 must be a digit, making O unambiguously 0. 7/7 pass rate on correction test suites.

3. Overlay Rejection: Detector filters out burnt-in timestamps, PTZ text overlays, and billboard signage before indexing.

Ground-Truth Benchmark: 6/6 plates recovered exactly, 0 false positives.

Live Sandbox Feed Findings: 29 of 30 cameras are wide-area PTZ overviews yielding only roadside signage (correctly rejected). **cam12 (Adalaj Toll Naka)** is a toll plaza feed: the pipeline detected a real truck plate **25 times in 100 seconds**, recovering 8 of 10 characters at 5 px/character. Seven preprocessing variants confirmed that upscaling cannot restore detail missing from optics.

The constraint is optics, not software (see Slide 10).

Slide 6: Continuous ANPR Engine & Scalability

Inference Costs & Siting Realities

At 1920×1080 resolution, wide-area night feeds render number plates at 5-15 px wide. Full-frame inference at 384 px proposes zero candidates. Tiled inference on overlapping upscaled regions recovers plate regions at 15x CPU cost.

Benchmark on cam05 (20-core CPU, No GPU)	Full Frame	Tiled 2x3 @2x
Mean Inference Time	12.3 ms/frame	185.9 ms/frame
Est. Concurrent Streams per Machine	≈251	≈ 26
Plate Candidates (90s window)	0	8
Ground-Truth Plates Recovered	3/6	6/6

Analytics Policy Tiers

ANPR is not a uniform per-camera cost. It is a **policy choice regarding which cameras require analytics**:

Analytics Tier	Share of Estate	Deployment Workload
Tier 1: Continuous ANPR	5-10%	Highways, border posts, ANPR-sited toll plazas
Tier 2: Event-Triggered	20-30%	Urban junctions (motion / signal triggered)
Tier 3: Registry & View	60-75%	Full video coverage without analytics compute cost

Slide 7: Correlation, Alerting & Route Reconstruction

Watchlist Matching: Exact + Fuzzy Trigram Search

Every detection is matched against active watchlists before database transaction commit: exact match first, followed by `pg_trgm` trigram similarity > 0.7. Matches generate instant WebSocket alerts with severity metadata.

```
SELECT *, similarity(plate_text, :q) AS score
FROM detections WHERE plate_text % :q
ORDER BY score DESC, detected_at;
```

Why Fuzzy Search is Critical: ANPR OCR will occasionally misread a character. Exact-only matching misses genuine detections. Verified: searching `GJ01AB1234` correctly retrieves records stored as `GJ01AB1234` .

Route Reconstruction

Sightings ordered chronologically → great-circle distance & elapsed time calculated → implied speed estimated → **physically impossible transitions (>150 km/h) flagged** and surfaced to operators rather than hidden.

Flagged transitions notify investigators of potential plate misreads or cloned registration plates, providing actionable intelligence.

Slide 8: Security, Privacy & Accountability

Security Control	Technical Implementation
Access Control	JWT authentication with 3 roles enforced per API endpoint
Audit Trail	Actor identity extracted from verified token with recorded purpose and case reference
Data Minimisation	Stores plate text, timestamp, camera, and plate crop (no raw video retention)
Data Residency	100% local inference: zero frames or crops leave local infrastructure
Credential Safety	Environment-based configuration, masked in logs, never sent to browser
Web Security	CSP, X-Frame-Options DENY, X-Content-Type-Options nosniff, HSTS
Rate Limiting	10 req/min on authentication, 300 req/min general (video streams exempt)
DPDP Act 2023	Purpose-bound queries, role restrictions, and auditability

Verified Enforcement: A viewer token receives **403 Forbidden** on audit trail and plate search routes, and **200 OK** on the camera registry.

Closed Government Systems (VAHAN, SARTHI, eGujCop)

Sentinel implements a contract-first adapter pattern. We defined request/response models, built mock adapters with realistic records, and documented exact activation steps when production credentials arrive (base URL, authentication, rate limits, audit hooks).

Slide 9: Scaling to 80,000 Cameras

Empirical Sandbox Ingestion Bottleneck Benchmark

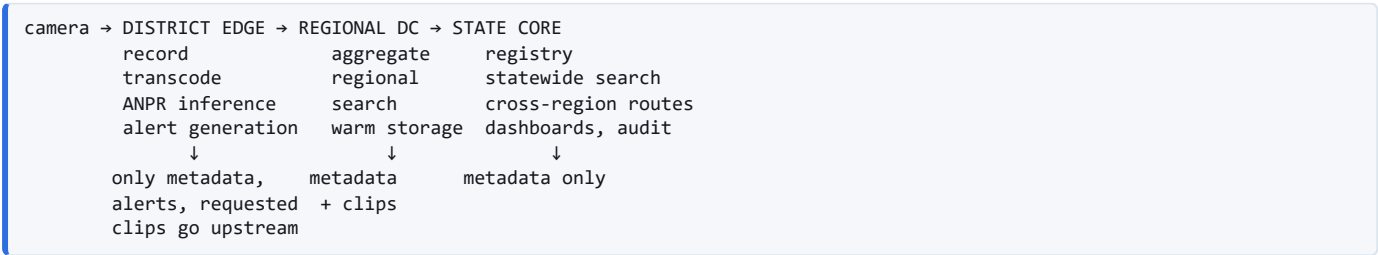
Opening N concurrent RTSP streams to the sandbox gateway:

Concurrent Streams	Succeeded	Wall Time	Our Server CPU (20 Cores)
2	2 / 2	12 s	4%
4	4 / 4	26 s	4%
6	6 / 6	41 s	5%
8	7 / 8	88 s	4%

Connection accept times for 8 streams: 4s, 8s, 27s, 33s, 56s, 60s, 63s, 73s (gateway queue buildup).

Key Insight: Our processing server remained idle at 4% CPU while the single gateway took 73s to accept 8 connections. Centralized single-ingress ingestion is the primary bottleneck.

Statewide Edge-First Architecture



Restricting WAN backhaul to metadata, alerts, and requested video clips reduces wide-area bandwidth demand by **2 to 3 orders of magnitude** (from 192 Gbps down to 8.7 Mbps).

Slide 10: Platform Capabilities & Transparent Evaluation

Fully Demonstrated Capabilities

- ✓ 30 sandbox cameras onboarded with full location provenance
- ✓ Interactive GIS map with department, status, and confidence filters
- ✓ Unified video wall featuring HLS grid and WebRTC hero tiles
- ✓ Continuous ANPR pipeline with tiled inference, track voting, and grammar correction
- ✓ Exact, partial, and fuzzy plate search
- ✓ Automatic watchlist matching with real-time WebSocket alerts
- ✓ Route reconstruction with speed calculation and impossible transition flagging
- ✓ Output report generation in XLSX and PDF formats
- ✓ Purpose-bound audit trail logging
- ✓ 57 automated unit/integration tests with 100% accuracy on legible ground-truth feeds

Transparent Limitations

- **Live Sandbox Feed Yield:** 29 of 30 cameras are wide-area PTZ overviews where plates measure 5-15 px (below optical resolution thresholds). cam12 (toll plaza) yielded 25 detections of a real truck plate at 5 px/character. Recommendation: prioritize toll plazas and checkpoints equipped with lane-facing optics.
 - **Geocoded Locations:** Camera coordinates are geocoded from site names, accurate to the facility rather than exact pole markers.
 - **Single-Node Deployment:** Prototype runs as a unified single-node deployment; scale-out topology is documented in HLD Section 9.
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Slide 11: Impact on Policing & Strategic Roadmap

Operational Transformation

- **Traditional Workflow:** Investigators manually contact multiple departments, scrub hours of footage, and assemble timelines over several days.
- **Sentinel Workflow:** Enter a vehicle registration once. Every sighting across all connected cameras appears on an interactive map within seconds.

Platform Capability	Operational Benefit
Continuous Indexing	Vehicle timelines exist prior to search queries
Automated Watchlist Alerts	Instant notifications when wanted vehicles pass cameras
Fuzzy Plate Matching	Misread characters do not prevent vehicle identification
Unified Camera Registry	Single access portal across departmental boundaries
Audited Access Control	Surveillance capabilities with complete accountability

Strategic Roadmap

1. **Phase 1:** Pilot deployment across 1 district (50-200 cameras) at toll plazas and checkpoints.
2. **Phase 2:** Multi-VMS adapter rollout across urban junctions (Model 3 integration).
3. **Phase 3:** Live VAHAN API credential activation and edge node expansion.
4. **Phase 4:** Statewide edge node deployment across 80,000 cameras (Model 4 scale-out).

Slide 12: Verification & Reproducibility

Claim / Benchmark	Verification Command / URL
ANPR Throughput Benchmark	<code>python anpr_worker.py --benchmark</code>
Ground-Truth 6/6 Accuracy Test	<code>python tools/make_sample_feed.py --validate</code>
Grammar & System Test Suite	<code>python -m pytest tests/ -q</code> (57 tests passing)
Ingest & Provenance Endpoint	<code>GET /api/v1/ingest/status</code>
Audit Trail API	<code>GET /api/v1/analytics/audit</code>
Interactive OpenAPI Docs	<code>http://localhost:8000/api/docs</code>